Supplementary Results

Survival-guided matrix factorization identifies reproducible prognostic programs in pancreatic cancer

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1 The DeSurv model shows consistent convergence and stability across restarts

Across datasets and initialization schemes, the DeSurv algorithm consistently converged to numerically stable solutions within the designated iteration budget. Warm-start strategies, in which solutions at $\alpha = 0$ initialized supervised runs, substantially reduced variability across restarts and improved reproducibility. Fig. S1 shows representative loss trajectories demonstrating monotone decreases until convergence across restarts. Compared with naive random initialization, warm-starts produced tighter distributions of cross-validated C-index and partial likelihood, confirming stability of the optimization procedure.

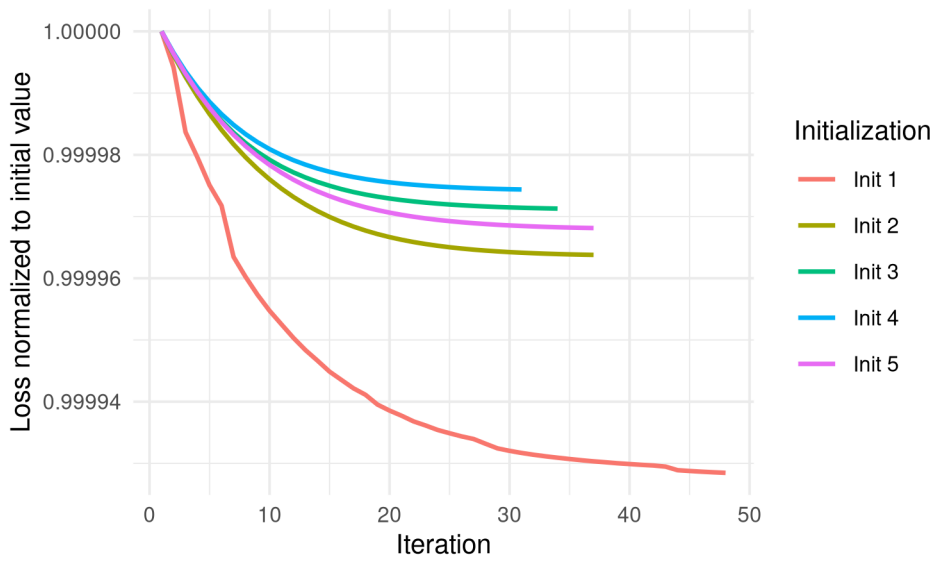

Figure S1: Convergence of model across initializations.

2 Simulation results under null and mixed scenarios

To characterize the operating characteristics of DeSurv across different signal regimes, we evaluated performance under three simulation scenarios. In the primary scenario (main text Fig. 3), prognostic programs explained low variance relative to outcome-neutral background signals, and survival depended on 150 factor-specific marker genes ($\beta_1 = 2$, $\beta_2 = \beta_3 = 0$). In the null scenario, survival times were generated independently of gene expression ($\beta = 0$), so no prognostic structure existed. In the mixed scenario, survival depended on 300 genes per lethal factor, of which 50% were factor-specific markers and 50% were background genes with loadings shared across all factors. All scenarios used $G = 3{,}000$ genes, $N = 200$ samples, $K = 3$ true factors, and identical gamma distribution parameters for the gene loading matrix W (see Materials and Methods). Performance was evaluated using cross-validated concordance index (C-index), precision of recovered prognostic gene sets, and the distribution of selected ranks across simulation replicates.

Under the null scenario (Fig. S2A–B), DeSurv showed no advantage over unsupervised NMF ($\alpha = 0$). Both methods yielded C-index values near 0.5, consistent with the absence of survival signal. The distribution of selected ranks was concentrated at low values for both methods, with NMF more strongly favoring $k = 2$. These results confirm that DeSurv does not hallucinate prognostic structure when none exists: Bayesian optimization selected low values of the supervision parameter α, effectively recovering standard NMF as a special case. This specificity control is important because it establishes that the advantages observed in the primary scenario reflect genuine signal recovery rather than overfitting to noise in the survival labels.

Under the mixed scenario (Fig. S2C–F), four mechanistic observations characterize DeSurv's advantage. First, when precision is evaluated against the factor-specific marker genes alone (150 genes; Fig. S2D left), DeSurv substantially outperforms NMF—matching or exceeding primary-scenario levels—even though overall survival-gene precision (300-gene survival set including background genes; Fig. S2D right) is more modestly improved. This dissociation reveals that background-gene recovery is similar between methods, and DeSurv's edge comes specifically from concentrating marker programs. Second, the learned factor most aligned with the true marker program receives a substantially larger Cox coefficient ($|\beta|$) under DeSurv than under NMF (Fig. S2E), confirming that survival supervision preferentially amplifies the marker-specific signal. Third, C-index values are modestly higher for DeSurv (Fig. S2C), consistent with the moderate overall improvement when background-gene dilution attenuates the survival signal. Together, these results demonstrate that DeSurv's advantage in the mixed scenario is mechanistic: the method specifically recovers marker programs and assigns them elevated survival weights, rather than incidental capture of background-gene prognosis.

Together, the three scenarios provide a dose-response relationship: DeSurv's advantage is greatest when variance and prognosis diverge (primary), moderate when they partially overlap (mixed), and absent when no survival signal exists (null). The mechanistic basis for the attenuation in the mixed scenario is now clear: DeSurv continues to specifically recover marker programs (high marker precision, elevated $|\beta|$), but the overall survival-gene precision improvement is dampened because background-gene dilution in the 300-gene survival signal reduces the discrimination advantage. This gradient is consistent with the predictions of sufficient dimension reduction theory and offers practical guidance for users: DeSurv is most beneficial in settings where the dominant transcriptional axes are expected to be prognostically neutral, as is common in cancers with high tumor purity variation or dominant tissue-composition signals.

3 Cross-validated C-index across factorization rank

To evaluate how the number of latent factors k affects prognostic performance, we performed an exhaustive grid search over $k \in \{2, 3, \ldots, 12\}$ and supervision strength $\alpha \in \{0, 0.05, \ldots, 0.95\}$ using 5-fold cross-validation on the TCGA+CPTAC training cohort. For each k, the best-performing α was selected by maximizing the mean CV C-index. The unsupervised NMF baseline ($\alpha = 0$) was evaluated at the same k values for comparison. Results are shown for two gene-selection settings: $n_{\text{top}} = 270$ top genes per factor and $n_{\text{top}} = \text{ALL}$ (all genes retained), both at $\lambda = 0.349$ and $\xi = 0.056$.

Fig. S3 shows the resulting C-index curves for both training (CV) and external validation (pooled across held-out cohorts), with shaded ribbons indicating ± 1 standard error. In training, DeSurv consistently outperforms NMF across all k, confirming that survival supervision improves in-sample concordance. In external validation, DeSurv at $k = 3$ achieves concordance comparable to NMF at higher ranks (e.g., $k = 7$; Table S4), demonstrating that survival supervision concentrates prognostic information into fewer factors rather than requiring additional rank to capture it.

4 Standard NMF rank selection heuristics

Standard unsupervised rank selection criteria — reconstruction residuals, cophenetic correlation coefficient, and mean silhouette width — yielded inconsistent guidance for selecting the factorization rank k in the PDAC training cohort (TCGA + CPTAC). Reconstruction residuals decreased smoothly without a clear elbow (Fig. S4A). Cophenetic correlation fluctuated without a distinct transition point (Fig. S4B). Mean silhouette width favored very low ranks that conflicted with other criteria (Fig. S4C). These contradictions motivated the survival-supervised rank selection approach described in the main text.

5 K-sensitivity analysis: effect of gene focusing on iCAF factor validation

To evaluate the robustness of the K=3 iCAF factorization, we conducted an exhaustive sensitivity analysis across $K \in \{2, 3, 5, 7, 9\}$ and $\alpha \in \{0, 0.25, 0.35, 0.55, 0.75, 0.85, 0.95\}$ (35 combinations), holding all remaining hyperparameters fixed at the production-optimal values ($\lambda = 0.349$, $\xi = 0.056$, 100 initializations). For each fit, we identified the factor most correlated with the production K=3 D1 (the iCAF factor) and evaluated its prognostic value in 570 independent validation patients across 4 cohorts (Dijk, Moffitt GEO, Puleo, PACA-AU), using stratified Cox regression (`strata(dataset)`) with and without adjustment for PurIST and DeCAF classifiers. Factor-program fidelity was quantified by Spearman rank correlation of $n_{\text{top}} = 270$-focused H-scores (H-cor) with the production factor. We repeated this analysis under two gene-projection settings, $n_{\text{top}} = 270$ (production-optimal top genes per factor) and $n_{\text{top}} = \text{NULL}$ (all 1,970 genes), to isolate the effect of gene focusing on validation performance.

The central finding is shown in Table S1: $n_{\textbf{top}} = 270$ **is the critical hyperparameter enabling the production fit's adjusted validation.** Moving from $n_{\text{top}} = \text{NULL}$ to $n_{\text{top}} = 270$, while holding λ and ξ at their optimal values, increases the number of K $\times$ α combinations achieving adjusted $P < 0.05$ from 5/35 to 18/35 and restores K=3 at $\alpha = 0.55$ to adjusted $P = 0.003$, nearly identical to the production fit ($P = 0.004$). By contrast, changing only λ from 0.3 to 0.349 while keeping $n_{\text{top}} = \text{NULL}$ leaves the overall count unchanged (5/35), confirming that λ/ξ tuning alone cannot reproduce the adjusted signal.

Table S1: Three-way comparison of K-sensitivity analyses differing only in gene-projection setting (n_{top}) and regularization (λ, ξ). All analyses span K $\in$ {2, 3, 5, 7, 9} $\times$ $\alpha \in$ {0, 0.25, 0.35, 0.55, 0.75, 0.85, 0.95} = 35 combinations. Count shows combinations achieving adjusted $P < 0.05$; bold highlights the $n_{\text{top}} = 270$ result that restores K=3 adjusted significance.

Analysis condition	Adj $P < 0.05$ (of 35)	K=3, α=0.55 adj P	Best single combination
n_{top} = NULL, $\lambda = 0.300$, $\xi = 0.050$	5 / 35	0.072	K=7, α=0.75 (adj $P = 0.019$)
n_{top} = NULL, $\lambda = 0.349$, $\xi = 0.056$	5 / 35	0.097	K=5, α=0.25 (adj $P = 0.0003$)
$n_{\text{top}} = 270$, $\lambda = 0.349$, $\xi = 0.056$	**18 / 35**	**0.003**	K=3, α=0.55 (adj $P = 0.003$)

Tables S2 and S3 present the full K $\times$ α adjusted P-value matrices for the $n_{\text{top}} = 270$ and $n_{\text{top}} = \text{NULL}$ analyses respectively. With $n_{\text{top}} = 270$, K=3 achieves adjusted significance at 5 of 7 α values tested ($\alpha \in \{0.00, 0.25, 0.55, 0.75, 0.95\}$), and broad adjusted significance is also observed at K=2 (5/7) and K=5 (4/7). With $n_{\text{top}} = \text{NULL}$ (Table S3), the landscape is sparse and structurally different: K=3 at $\alpha = 0.55$ narrowly misses significance ($P = 0.097$), and only isolated combinations scattered across K=3, K=5, K=7, and K=9 achieve adjusted $P < 0.05$.

Table S2: Adjusted validation P-values (Cox + PurIST + DeCAF + strata(dataset)) across K $\times$ α with $n_{\text{top}} = 270$, $\lambda = 0.349$, $\xi = 0.056$. Each cell gives the P-value for the factor most correlated with the production K=3 iCAF factor. Bold: $P < 0.05$; 18 of 35 combinations are significant.

	α						
	0.00	0.25	0.35	0.55	0.75	0.85	0.95
K=2	0.564	**0.038**	**0.038**	**0.019**	0.060	**0.011**	**0.002**
K=3	**0.030**	**0.030**	0.519	**0.003**	**0.028**	0.263	**0.009**
K=5	**5e-4**	**0.005**	**8e-4**	**0.019**	0.073	0.604	0.447
K=7	0.513	0.471	**6e-4**	**0.041**	0.676	0.131	0.130
K=9	**3e-4**	0.299	0.243	**0.005**	0.943	0.867	0.263

Four additional findings emerge from these analyses. First, K=7's apparent advantage observed in a parallel analysis using conventional hyperparameters ($\lambda = 0.3$, $n_{\text{top}} = \text{NULL}$), where K=7 at $\alpha = 0.75$ achieved

Table S3: Adjusted validation P-values across K $\times$ α with n_{top}=NULL (all 1970 genes), $\lambda = 0.349$, $\xi = 0.056$. Layout identical to Table S2. Only 5 of 35 combinations are significant (bold), and K=3 at $\alpha = 0.55$ (the production neighborhood) reaches only $P = 0.097$.

	α						
	0.00	0.25	0.35	0.55	0.75	0.85	0.95
K=2	0.414	0.192	0.142	0.197	0.090	0.194	0.158
K=3	0.205	**0.024**	0.280	0.097	0.112	0.076	0.111
K=5	0.461	**3e-4**	0.150	0.233	0.067	0.124	0.197
K=7	0.394	0.336	0.304	0.097	0.094	0.062	**0.021**
K=9	**0.023**	0.093	0.474	**0.022**	0.106	0.127	0.122

adjusted $P = 0.019$, completely disappears with $n_{\text{top}} = 270$ (adj $P = 0.676$, Table S2). Focused 270-gene projection fragments the iCAF signal across K=7's factors, indicating that the earlier K=7 advantage depended on the all-genes projection rather than reflecting an intrinsic property of the 7-factor solution. Second, K=3 achieves adjusted significance across a broad α range with $n_{\text{top}} = 270$ (5/7 values), demonstrating that the production result does not depend on the BO-selected $\alpha = 0.334$ specifically. Third, K=5 at $\alpha = 0.55$ achieves the highest factor-program fidelity in the $n_{\text{top}} = 270$ analysis (H-cor = 0.903 vs. K=3's 0.855) and also achieves adjusted significance ($P = 0.019$), providing convergent evidence for the iCAF biology from a higher-K model; an independent K=5 all-genes fit at $\alpha = 0.25$ likewise achieves adjusted significance ($P = 0.0003$, H-cor = 0.91). Fourth, without gene focusing, high supervision weights ($\alpha \geq 0.75$) produce factors with extreme training discrimination (HR $\approx$ 0.07–0.1, LRT $P \sim 1e-60$) that fail external validation, indicating overfitting of the training survival signal; gene focusing ($n_{\text{top}} = 270$) suppresses this by constraining each factor to its 270 most biologically characteristic genes.

6 K=3 factors are partially but asymmetrically nested within K=7

To assess whether the K=3 solution is simply a pruned version of the K=7 solution, we computed pairwise Spearman correlations between the W-matrix gene loadings of the production K=3 model and a K=7 DeSurv fit at $\alpha = 0.35$ (the nearest grid value to the BO-selected optimum at $\alpha = 0.334$), both using $n_{\text{top}} = 270$, $\lambda = 0.349$, $\xi = 0.056$.

The correspondence was partial but asymmetric (Fig. S5). Two of three K=3 factors had clear one-to-one K=7 counterparts: K3_F1 mapped strongly to K7_F1 ($r = 0.77$) and K3_F2 mapped very strongly to K7_F7 ($r = 0.88$). In contrast, K3_F3 showed no single dominant K=7 match; instead, it correlated moderately with four K=7 factors simultaneously (K7_F2 through K7_F6; $r = 0.57$–0.71). K7_F4 showed weak-to-moderate correlations with all three K=3 factors ($r = 0.45$–0.52), suggesting it is a mixed composite with no clear biological counterpart in the K=3 solution.

This fragmentation pattern indicates that K=7 does not simply subdivide K=3 structure. Rather, it disperses one coherent K=3 prognostic program across multiple lower-signal components, consistent with the observed failure of K=7 to achieve adjusted significance in external cohorts (adjusted $P = 0.676$; see main text).

7 Cutpoint selection and validation of dichotomized risk groups

To convert the continuous DeSurv linear predictor into a clinically interpretable binary risk stratification, we selected an optimal z-score cutpoint via cross-validation on the TCGA+CPTAC training cohort. For each candidate cutpoint in $\{-2.0, -1.8, \ldots, 2.0\}$, we computed the mean absolute log-rank z-statistic across held-out folds and selected the cutpoint maximizing this metric (Fig. S6A). We then applied this cutpoint to external validation cohorts by standardizing each patient's linear predictor using the training mean and

standard deviation. Fig. S6B shows the resulting Kaplan–Meier curves for the pooled external validation analysis, with a stratified log-rank test accounting for dataset of origin. Panels C–F display per-cohort validation KM curves for Dijk, Moffitt, PACA-AU, and Puleo, respectively. Across all cohorts, the DeSurv-derived high-risk group showed consistently shorter survival, confirming that the training-derived cutpoint generalizes to independent datasets.

8 Overlap of DeSurv risk groups with known molecular subtypes

To assess whether the DeSurv-derived risk stratification captures biologically meaningful transcriptional programs, we examined the overlap between dichotomized risk groups (High vs Low) and two established PDAC molecular classifiers: PurIST (Basal-like vs Classical) and DeCAF (proCAF vs restCAF). Fig. S7 shows the composition of each risk group across all pooled external validation cohorts, with Fisher's exact test P-values quantifying the association. The high-risk group was significantly enriched for Basal-like (PurIST) and proCAF (DeCAF) subtypes, consistent with the known poor prognosis of these molecular classes. This concordance confirms that DeSurv's survival-driven factorization recovers clinically relevant biology without requiring subtype labels during training.

9 Standard NMF factor structure at independently selected rank

Throughout the main text, both DeSurv and standard NMF were fit at the same rank ($k = 3$), enabling a controlled comparison of the effect of survival supervision on factor structure. When each method selects its own rank independently, standard NMF chooses $k = 5$ via residual elbow detection or $k = 7$ via Bayesian optimization with $\alpha = 0$, while supervised DeSurv BO selects $k = 3$. We examine here the gene program structure at elbow-selected $k = 5$ for standard NMF (Fig. S8), and the NMF $k = 7$ structure to illustrate additional fragmentation when supervision is absent (Fig. S9).

At elbow-selected $k = 5$, the three core programs from $k = 3$ are recovered by standard NMF (Fig. S8). Two additional factors emerge with strong overlap with Exocrine-like and iCAF signatures, capturing transcriptional variance not aligned with survival outcomes. These biologically redundant factors explain why supervised BO selects $k = 3$ rather than $k = 5$: the additional rank adds no prognostically relevant structure.

At BO-selected $k = 7$ under $\alpha = 0$ (Fig. S9), standard NMF shows increased fragmentation relative to $k = 5$: multiple factors partially overlap the same PDAC gene signatures, and no new biologically coherent program emerges beyond those present at $k = 3$. A supervised DeSurv equivalent at $k = 7$ is not shown because supervised BO selects $k = 3$, not $k = 7$; the relationship between the production $k = 3$ DeSurv model and a $k = 7$ supervised DeSurv fit ($\alpha = 0.35$) is characterized by the factor-nesting analysis (Fig. S5), which demonstrates that the $k = 7$ solution fragments rather than extends the $k = 3$ structure.

Table S4: External validation C-index by method and cohort. DeSurv at BO-selected rank ($k = 3$) is compared with standard NMF at matched rank ($k = 3$), elbow-selected rank ($k = 5$), and BO-selected rank ($k = 7$, $\alpha = 0$). Higher values indicate better discrimination.

Cohort	DeSurv (K=3)	Std NMF (K=3)	Std NMF (K=5, elbow)	Std NMF (K=7, BO α=0)
Dijk	0.607	0.541	0.633	0.631
Moffitt_GEO_array	0.562	0.521	0.550	0.579
PACA_AU_array	0.649	0.470	0.640	0.657
PACA_AU_seq	0.634	0.430	0.674	0.695
Puleo_array	0.636	0.534	0.640	0.652

Standard NMF at elbow-selected $k = 5$ and BO-selected $k = 7$ both substantially improve over NMF at $k = 3$ across all cohorts (Table S4), confirming that additional unsupervised factors capture prognostically relevant variation missed at the matched rank. At $k = 7$, NMF matches or exceeds DeSurv's per-cohort C-index in most cohorts, demonstrating that unsupervised factorization can approach supervised concordance given enough factors. However, this comes at the cost of a less parsimonious and less interpretable factorization: the $k = 7$ NMF solution shows increased fragmentation (Fig. S9), and the K-sensitivity analysis (Tables S1–S3) shows that $k = 7$'s prognostic signal largely overlaps with existing molecular classifiers (PurIST, DeCAF), whereas $k = 3$ DeSurv retains significance after classifier adjustment. DeSurv's advantage thus lies not in maximal concordance but in recovering a compact, interpretable factorization whose prognostic content is independent of known classifiers.

To quantify this directly, Table S5 reports pooled validation hazard ratios (per SD of the linear predictor) for DeSurv at $k = 3$ and standard NMF at BO-selected $k = 7$, both unadjusted and after adjustment for PurIST and DeCAF molecular classifiers. In the unadjusted analysis, both methods show a survival association, though DeSurv's effect size is substantially larger. After adjustment, DeSurv retains a significant and clinically meaningful hazard ratio, whereas NMF's effect attenuates and is no longer significant, confirming that NMF's prognostic content at $k = 7$ is largely redundant with existing classifiers.

Table S5: Pooled external validation hazard ratios (per SD of linear predictor) for DeSurv at $k = 3$ and standard NMF at BO-selected $k = 7$ ($\alpha = 0$). Adjusted models include PurIST and DeCAF molecular classifiers as covariates, with stratification by dataset.

Method	Unadjusted HR per SD (95% CI), P	Adjusted HR per SD (95% CI), P
DeSurv ($k = 3$)	1.45 (1.29–1.63), < 0.001	1.3 (1.13–1.5), < 0.001
Std NMF ($k = 7$, BO $\alpha = 0$)	1.46 (1.32–1.61), < 0.001	1.42 (1.23–1.64), < 0.001

The variance–survival decomposition at $k = 5$ (Fig. S10) confirms that elbow selection does not recover additional prognostic structure. At $k = 5$, standard NMF continues to distribute survival contribution negligibly across all factors; the two factors new relative to $k = 3$ (N4 and N5) contribute minimally, confirming that the additional rank captures background transcriptional variance rather than independent prognostic programs.

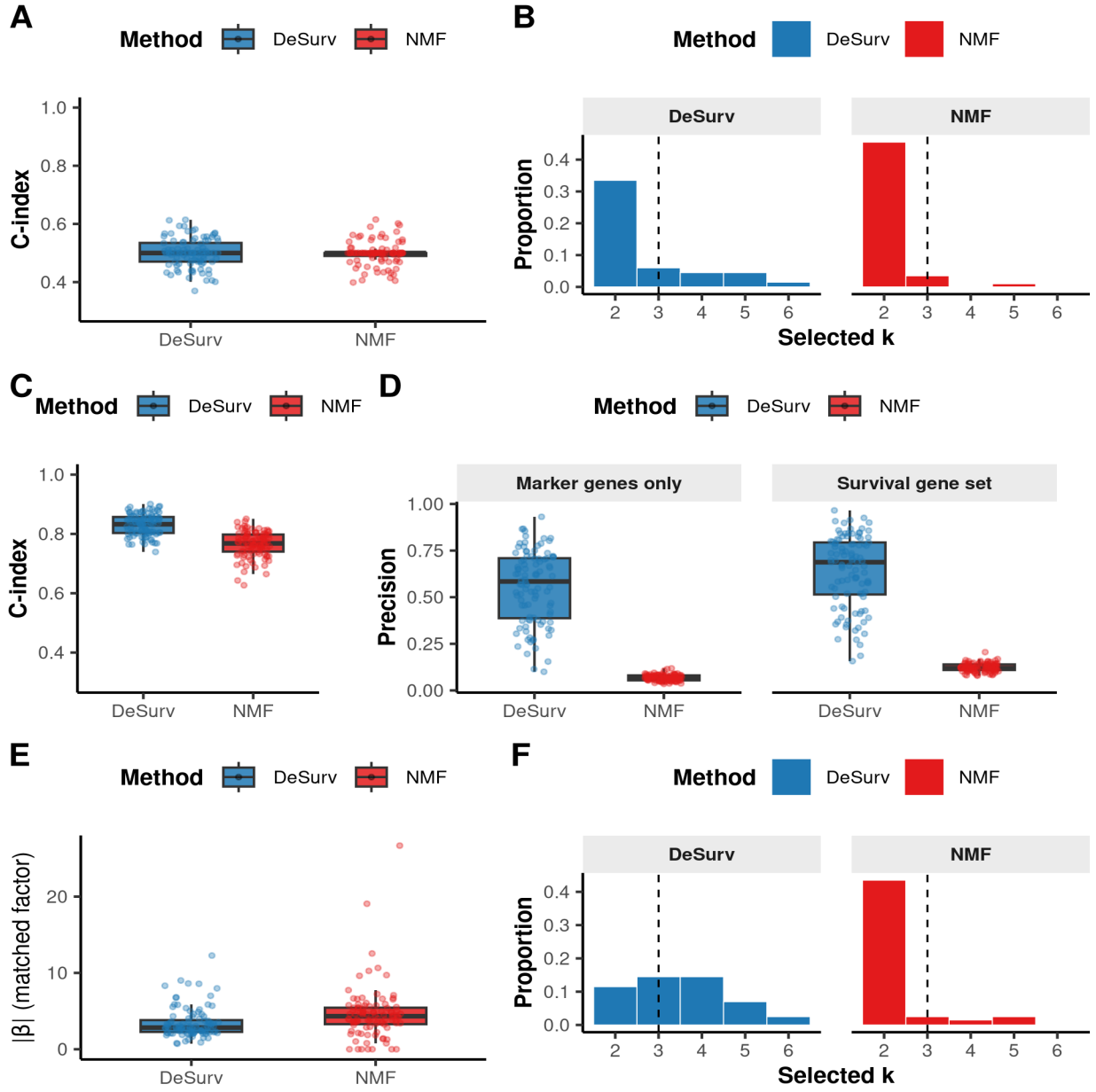

Figure S2: DeSurv performance under null and mixed simulation scenarios. (A–B) Null scenario ($\beta = 0$): both methods yield C-index ≈ 0.5 (A) and comparable rank distributions (B), confirming DeSurv does not hallucinate prognostic structure; precision is omitted as no true survival genes exist. (C–F) Mixed scenario: survival depends on factor-specific marker genes (50%) and shared background genes (50%). (C) C-index is modestly higher for DeSurv. (D) Precision by gene-set definition: marker-only precision (150 genes; left facet) substantially favors DeSurv; full 300-gene survival-set precision (right facet) shows a more modest improvement. (E) The matched factor receives a substantially larger Cox coefficient $|\beta|$ under DeSurv, confirming preferential amplification of marker-specific signal. (F) Selected-rank distributions. DeSurv (blue); NMF (red, $\alpha = 0$); both Bayesian-optimized.

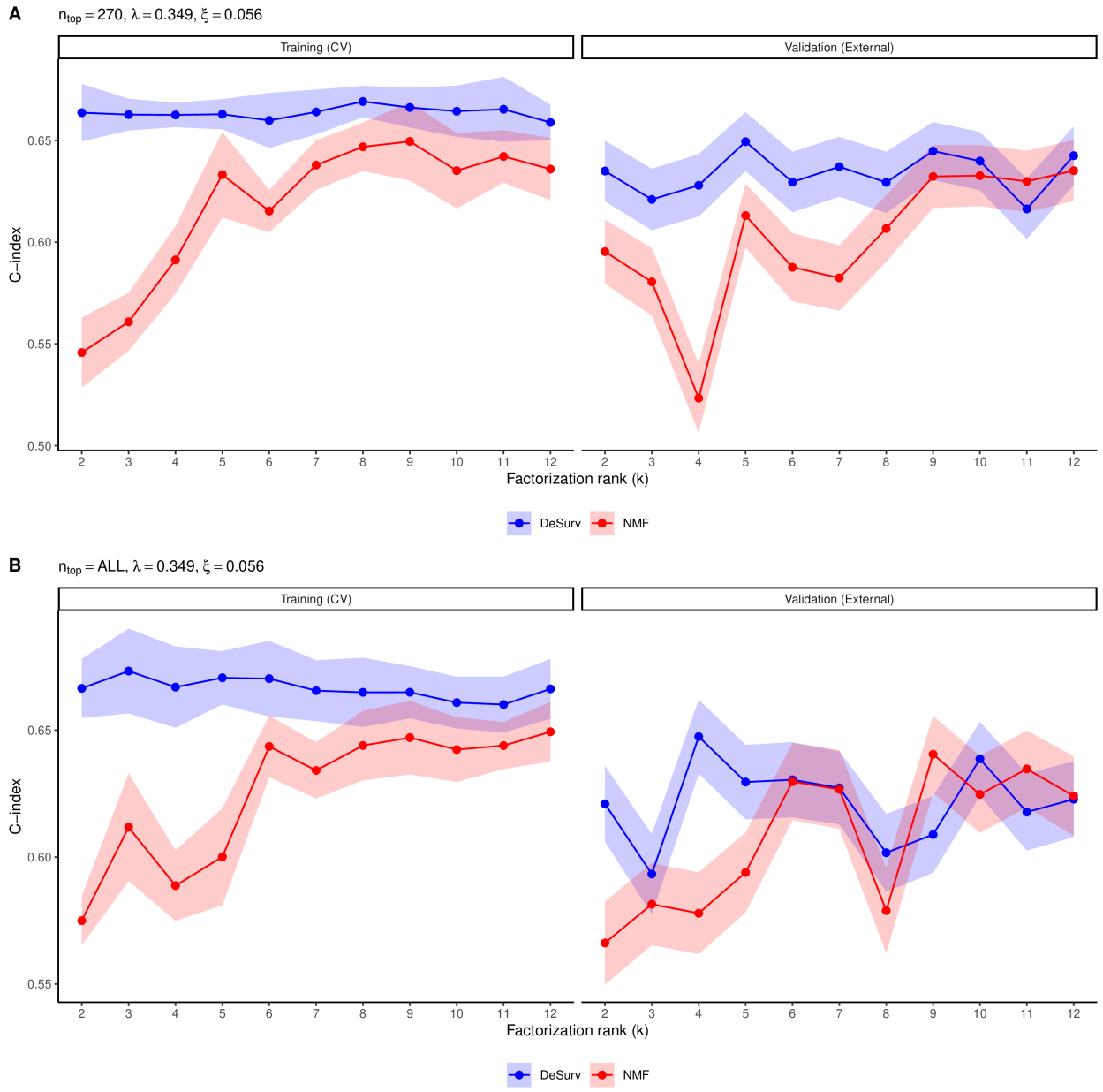

Figure S3: Training (CV) and external validation C-index as a function of factorization rank k for DeSurv (blue) and standard NMF (red, $\alpha = 0$). Shaded ribbons show ± 1 SE. For each k, the best α was selected by maximizing the mean CV C-index. Validation C-index is pooled across all external cohorts. (A) Results using $n_{\text{top}} = 270$ top genes per factor ($\lambda = 0.349$, $\xi = 0.056$). (B) Results retaining all genes ($n_{\text{top}} = \text{ALL}$; $\lambda = 0.349$, $\xi = 0.056$).

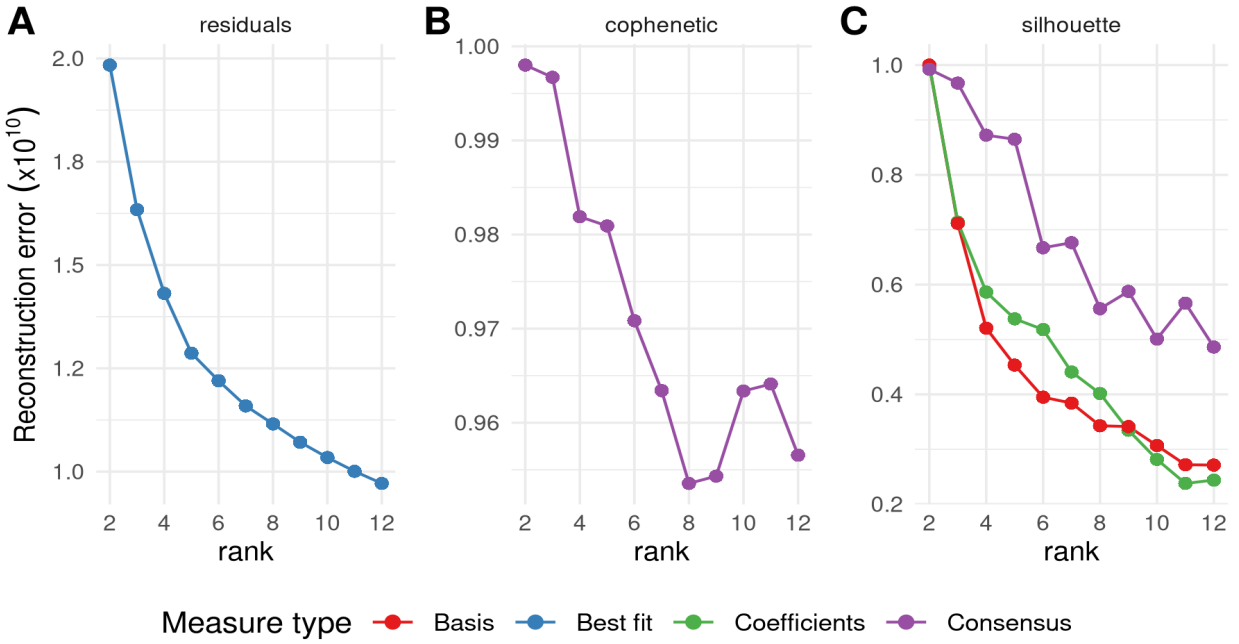

Figure S4: Standard unsupervised NMF rank selection heuristics for the PDAC training cohort (TCGA + CPTAC). (A) Reconstruction residuals as a function of factorization rank k. (B) Cophenetic correlation coefficient as a function of k. (C) Mean silhouette width across multiple distance metrics as a function of k. These criteria yield inconsistent guidance, motivating the survival-supervised model selection approach (main text Fig. 2D).

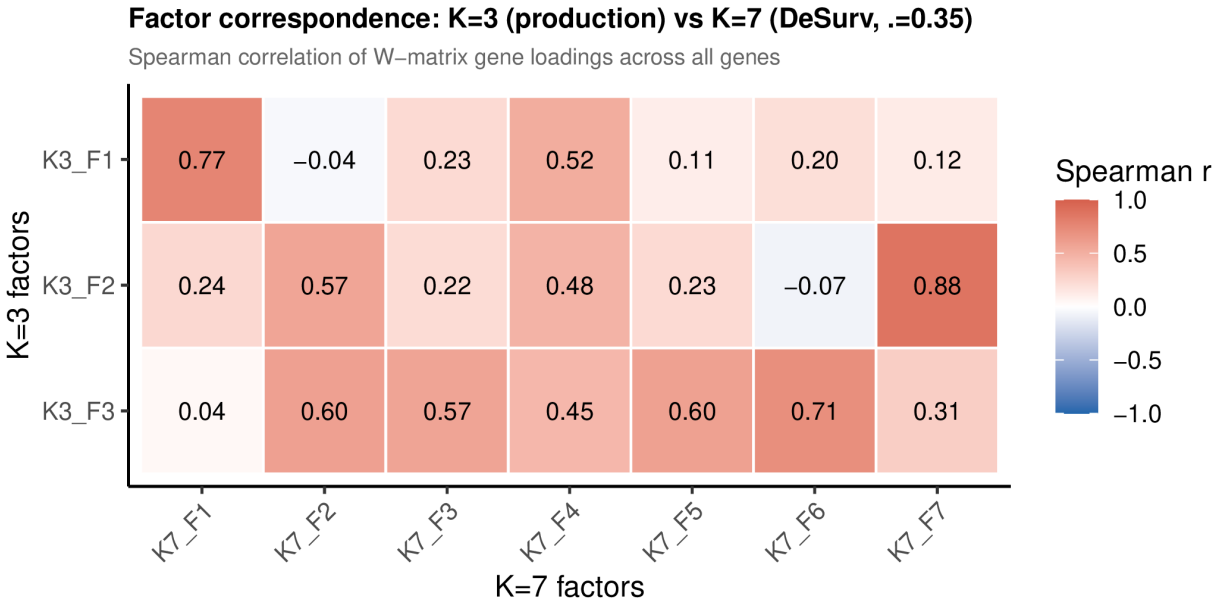

Figure S5: Factor correspondence between the production K=3 model and a K=7 DeSurv fit ($\alpha = 0.35$, $n_{\text{top}} = 270$), measured by Spearman rank correlation of W-matrix gene loadings across all genes. Two K=3 factors have clear K=7 counterparts (K3_F1 → K7_F1, $r = 0.77$; K3_F2 → K7_F7, $r = 0.88$), while K3_F3 is distributed broadly across four K=7 factors ($r = 0.57$–0.71), indicating fragmentation rather than nesting.

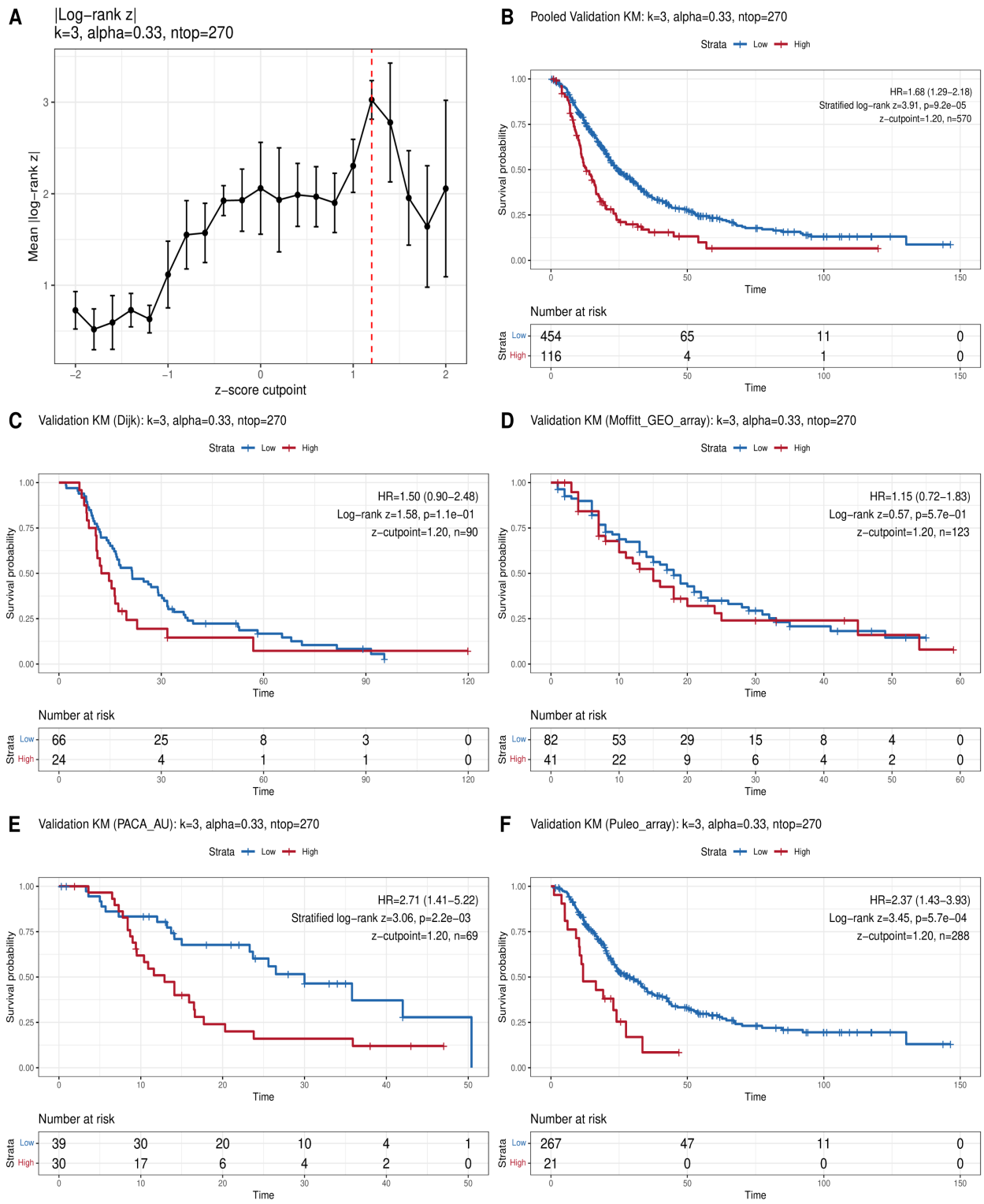

Figure S6: Cutpoint selection and external validation of dichotomized risk groups. (A) Cross-validated mean absolute log-rank z-statistic as a function of z-score cutpoint; red dashed line indicates the selected optimum. (B) Pooled validation Kaplan–Meier curves (stratified log-rank test). (C–D) Per-cohort validation KM curves: (C) Dijk, (D) Moffitt. (E–F) Per-cohort validation KM curves: (E) PACA-AU, (F) Puleo.

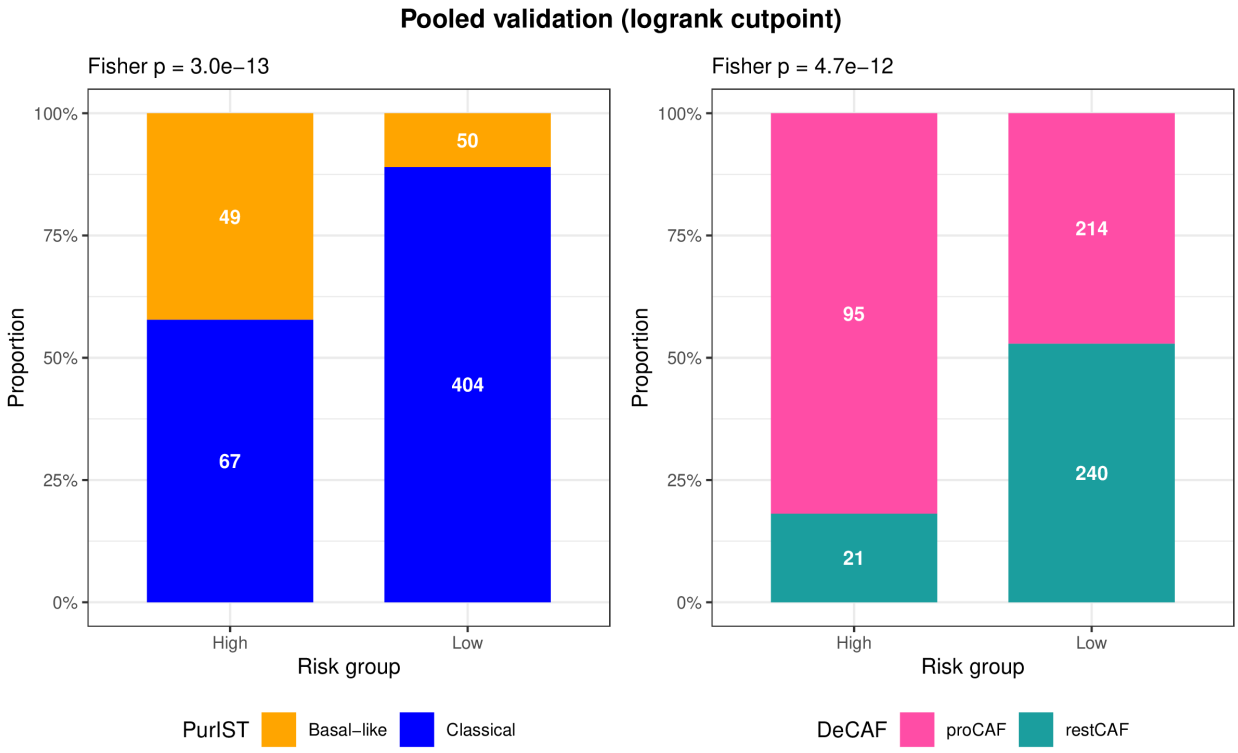

Figure S7: Subtype composition of DeSurv risk groups in pooled external validation cohorts. Stacked bar charts show the proportion of PurIST (left) and DeCAF (right) subtypes within Low and High risk groups defined by the log-rank-optimized cutpoint. Numbers indicate sample counts; Fisher's exact test P-values are shown above each panel.

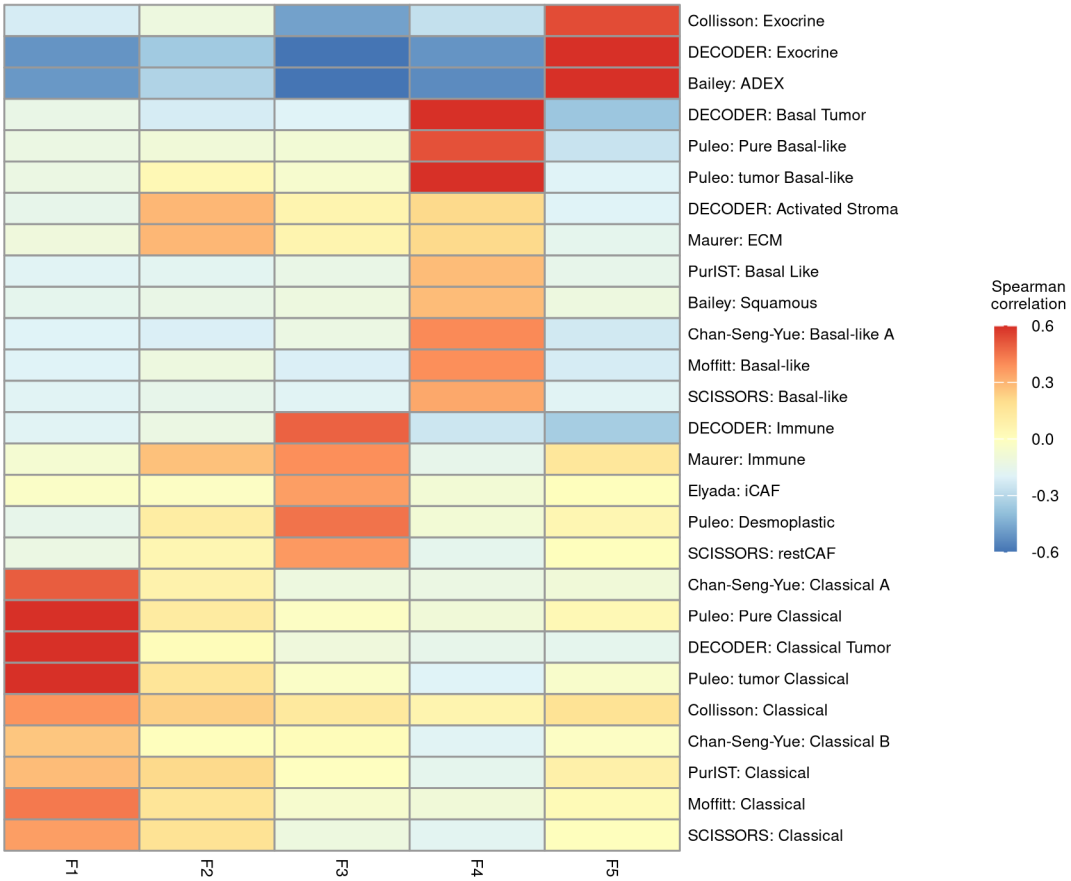

Figure S8: Gene program overlap heatmap for standard NMF at elbow-selected rank $k = 5$. Each cell shows the number of top genes from a given factor (rows) overlapping established PDAC gene programs (columns). The three core programs from $k = 3$ are recovered; two additional factors overlap Exocrine-like and iCAF signatures, capturing transcriptional variance not aligned with survival outcomes.

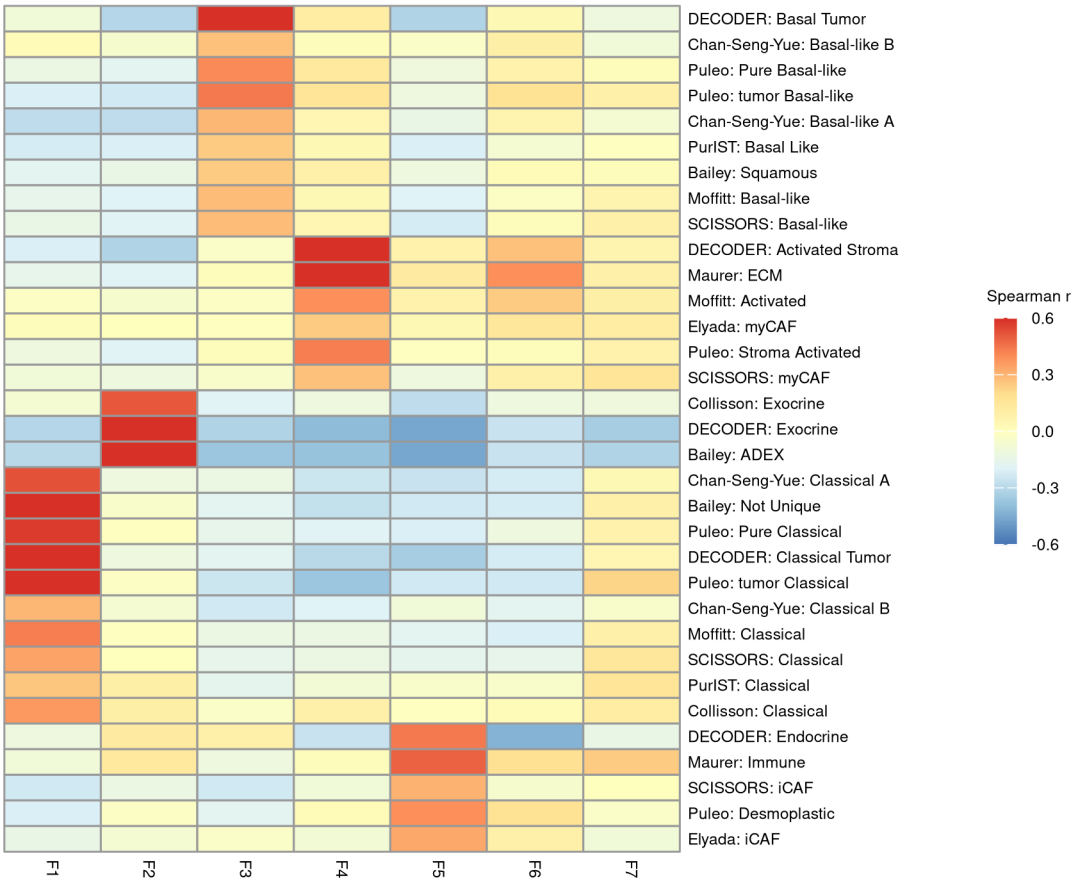

Figure S9: Gene program overlap heatmap for standard NMF at $k = 7$ (BO-selected rank with $\alpha = 0$). Compared with $k = 3$ and $k = 5$, the 7-factor solution shows increased fragmentation: multiple factors partially capture the same PDAC gene signatures and no new coherent biological program emerges. No supervised DeSurv equivalent at $k = 7$ is shown; the k-sensitivity and factor-nesting analyses (Fig. S5) characterize the corresponding supervised fit.

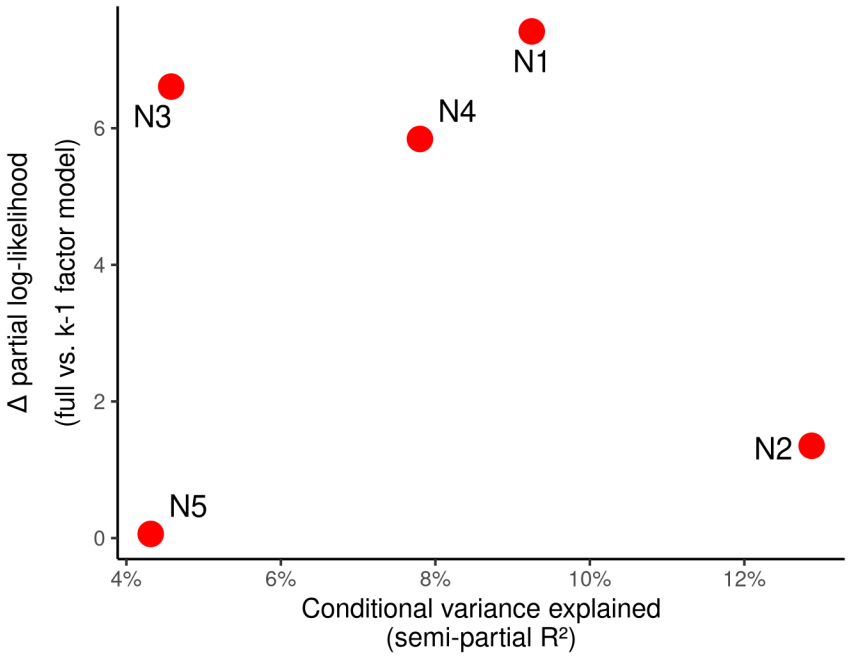

Figure S10: Fraction of expression variance explained versus Type III partial likelihood contribution for each standard NMF factor at elbow-selected $k = 5$. Each point represents one factor, labeled N1–N5. Survival contribution is quantified as the reduction in Cox partial log-likelihood when each factor is removed from the full model. The two factors new relative to $k = 3$ (N4, N5) contribute negligibly to survival, confirming that elbow selection recovers no additional prognostic structure.